

Devarpana Tribedi

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Professional Summary

BSc (Hons.) Data Science and Artificial Intelligence student at IIT Guwahati with hands-on experience in Computer Vision, data annotation, data processing, IoT systems, and web development through internship projects at TIH-IITG. Worked on real-world railway applications involving computer vision, IoT, and data-driven web applications, with additional hands-on experience fine-tuning YOLOv8 for PPE detection.

Education

Indian Institute of Technology Guwahati

2023 – 2027

BSc (Hons.) in Data Science and Artificial Intelligence

CGPA: 7.87

Technical Skills

Programming: Python, C, C++, Java, R, JavaScript

Machine Learning & Computer Vision: Computer Vision, YOLOv8, Object Detection, Model Fine-tuning, Data Annotation, Data Collection, Data Filtering, Data Preprocessing

Data & Databases: Pandas, NumPy, Matplotlib, MongoDB, MySQL, Data Visualization

Web Development: HTML, CSS, JavaScript, React.js, JSX, Tailwind CSS, Node.js, Express.js, Django

Experience

Technology Innovation Hub (TIH), IIT Guwahati Technology Innovation & Development Foundation (IITG-TIDF)

Nov 2025 – July 2026

Computer Vision Intern — Web Developer Intern

Railway Door Lock Detection System

- Annotated approximately 10,000 train images for door and door-lock detection using Roboflow.
- Performed manual data filtering and quality control by removing noisy images and retaining samples with clearly visible train doors.
- Developed a multi-user web application using Django, React, JSX, and Python to showcase railway data from the server.

Railway Chain Pull Alarm System

Live Website

- Worked on an ESP32-based IoT system for detecting chain-pull events in running trains and uploading event data to a server.
- Worked with ESP32, GPS/GSM, and optocoupler components using C++ for device-side development.
- Handled chain-pull event data across approximately 60 coaches in 3 trains and supported web-based presentation of server data using MongoDB.
- Live Website

Projects

Helmet/PPE Detection — YOLOv8

- Fine-tuned a pretrained YOLOv8 object detection model on approximately 7,000 Roboflow images to detect helmet, PPE, and person classes.
- Implemented image-based inference to detect and localize helmet, PPE, and person objects using bounding-box predictions.

Take Up Education Solutions — Web Development

Live Website

- Developed a production-ready study-abroad consulting website using React, Next.js, V0, Tailwind CSS, HTML, CSS, and JavaScript.
- Built responsive interfaces and interactive features including program browsing, university filters, and contact forms.
- Optimized the website for performance, responsiveness, and consistent user experience across desktop and mobile devices.

Certifications

- Computer Vision Internship Certificate – TIH-IITG / IITG-TIDF
- Web Developer Internship Certificate – TIH-IITG / IITG-TIDF
- Oracle Cloud Infrastructure 2025 – Certified AI Foundations Associate
- Oracle Data Platform 2025 – Certified Foundations Associate
- Generative AI Mastermind – Certificate of Completion, Outskill

Relevant Coursework

Machine Learning, Deep Learning, Computer Vision, Linear Algebra, Calculus, Optimization, Probability, Statistics, Data Mining, Data Visualization, Data Structures and Algorithms, RDBMS, Data Modeling, Cloud Computing